Name: A Solomon

Topic; Diagnostic Analysis of the NHS Using Python

02-March-2026

ABSTRACT

This report seeks to explore and explain capacity and resource utilisation issues confronting the NHS. Both internal and external data sources were utilised, total of four datasets. Findings include evidence of optimum capacity and resource utilisation, vast number of appointments are attended, mostly within 1 -7 days with about 100,000 missed appointments per month. And the system is GP led. The most dominant topic for discussion within the population is #healthcare which will send a signal to policy makers.

Background and Context

This analytical report having the nature of exploratory and diagnostic analysis concerns the NHS, a publicly owned healthcare service. This report is for the management team at the NHS who will make decisions based on the trends and insights produced in this report.

The Problem Statement

The NHS management are interested in understanding questions in three (3) broad areas:

- Utilisation of resources
- Missed appointments
- The potential value of utilising social media, Tweeter (now 'X') in this analysis

These broad questions served as a guide to formulating specific and relevant questions that will be helpful to the stakeholders. These are underpinned by the rationale that as the population keeps increasing, and people living longer, pressures on the healthcare services shall become increasingly strained. In the area of missed appointment, this is a lost time which translate into wasted resources. With regard to the potential of Tweeter, this offers valuable information that can direct healthcare policy.

The Analytic Approach

This section outlines the data sources and the methodology adopted to analyse the dataset. It also utilised both exploratory and explanatory data analysis.

Data Source: In this report, two (2) main sources of data were utilised: Internal & External. The internal data comprised of three (3): actual_duration.csv, appointments_regional.csv and national_categories.xlsx and a complementary text in the form of metadata (metadata_nhs.text). The external data is obtained from Tweeter (tweets.csv).

Methodology: The above data sources were imported into Jupyter Notebook by Pandas and appropriate python code snippets were used to store these data in DataFrames for ease of analysis. The DataFrames were then analysed for trends and insights using python code snippets. The code cells were annotated with the appropriate comments and markdown explanations. The relevant outputs from the python code snippets are then visualised using seaborn and matplotlib to aid in identifying trends and insights contained within the dataset. Some of the chart's types and the rationale behind their use include:

- Lineplots – used to show trends over time. As the stakeholders are interested in determining how the system is performing across the relevant months of the study.
- Barplots – used to show comparison between categories. With this the stakeholders can compare healthcare professionals are leading the services and which service settings are mostly preferred by users. Also, which hashtags get retweeted and liked/favourited to be termed as trending.
- Boxplot – used to show spread and outliers. Even though technical stakeholders are mostly interested in boxplots, the business stakeholders of the NHS can also benefit from the insight that boxplots offer. Here, the NHS bosses can know which services or locations are extremely overloaded or completely underperforming which can inform decisions on resource reallocation.

How the Data Helped to Answer the Questions: There seemed to be minimal data quality issues within the dataset. But efforts were made to check for duplications and missing data. One issue identified is the inconsistency in the date ranges among the datasets. To align the date ranges, the start off month was adjusted to August 2021. Having completed quality check, the dataset contained enough information to answer the broad three questions mentioned earlier and a much more specific business questions facing the NHS. Some of these include:

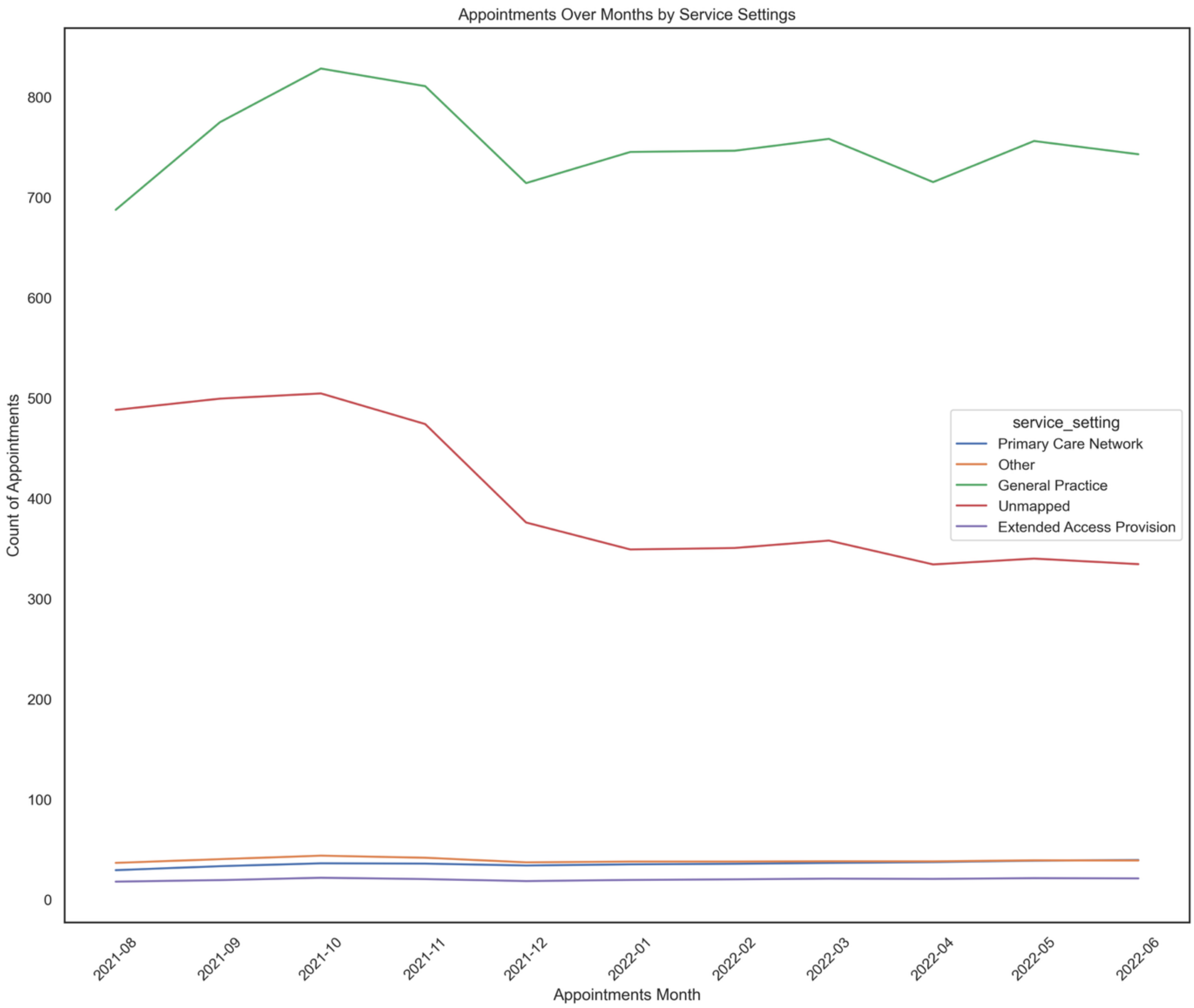
- How many types of service settings and which is most preferred
- How many appointments are booked per month
- How many appointments are attended per day
- How long does it take from date of booking to attending an appointment
- Are there seasonal variations in service demands and which services are in high demands
- Are there enough resources such as staff to cater for the increasing needs.
- What healthcare topics are being discussed among the population that stakeholders need to take into consideration. Etc.

Visualisation and Insights

This section puts together the findings of the analysis in the form of charts and tables. This report used exploratory analysis to explore/investigate the dataset to by asking specific questions for specific answers such as number of appointments per month. By contrast, explanatory analysis seeks to communicate key findings to support decision making. Both methods were used.

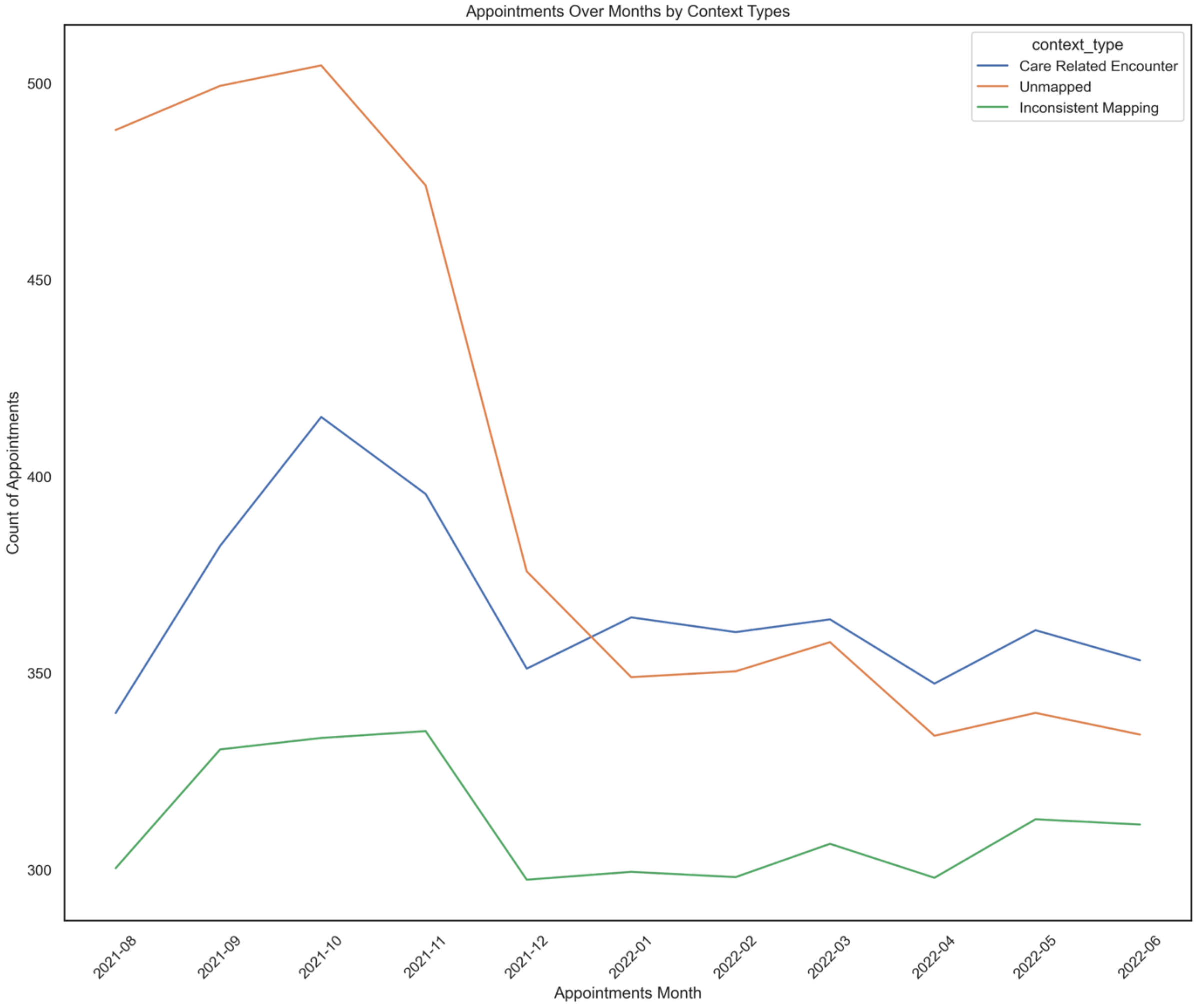
Below are some of the key findings in the form of charts:

Figure 1.



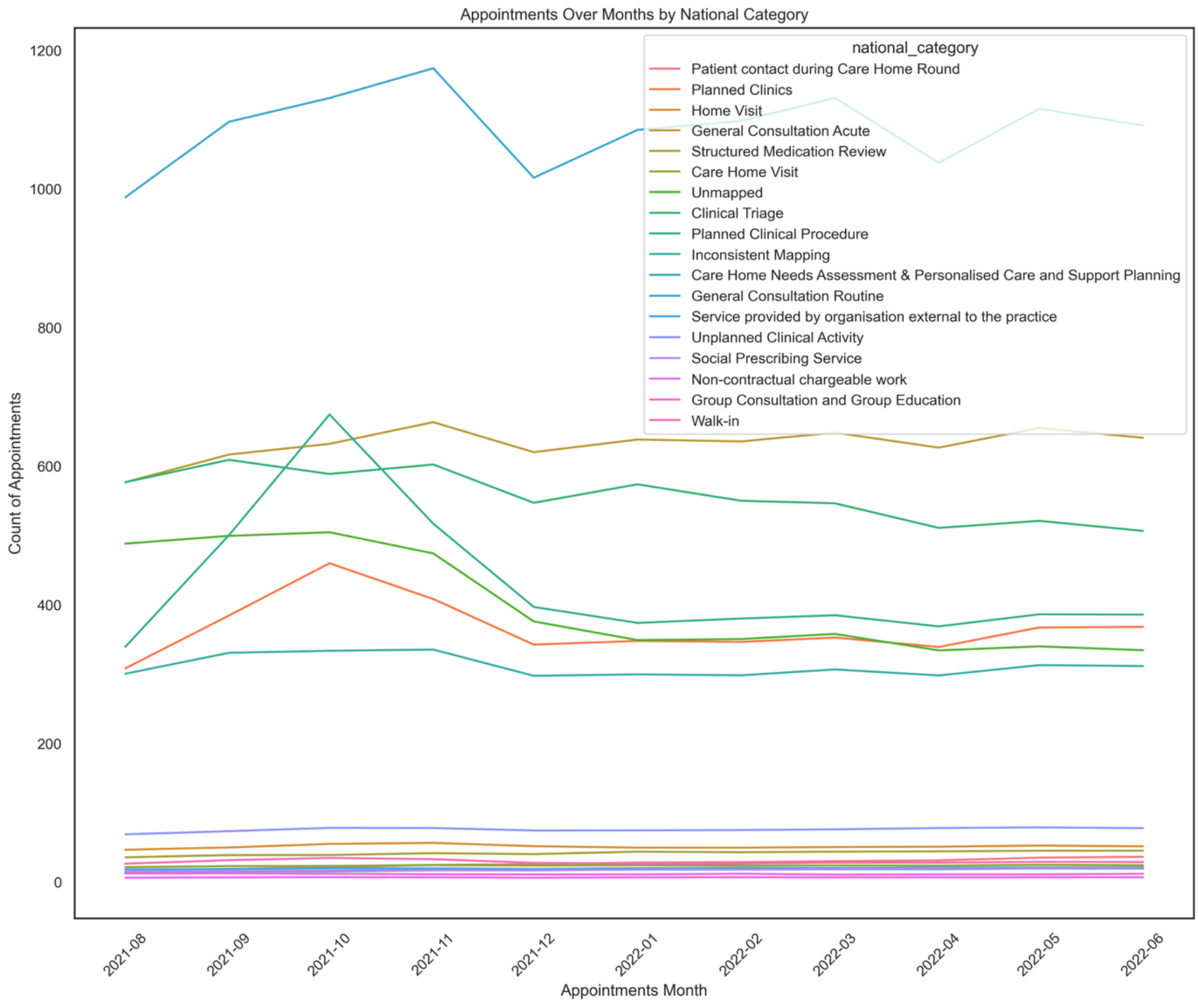
This lineplot shows the distribution of appointments across the various service settings. This chart shows that, General Practice (GP) heavily dominates how healthcare is delivered. Key insight is that, NHS services are GP led.

Figure 2.



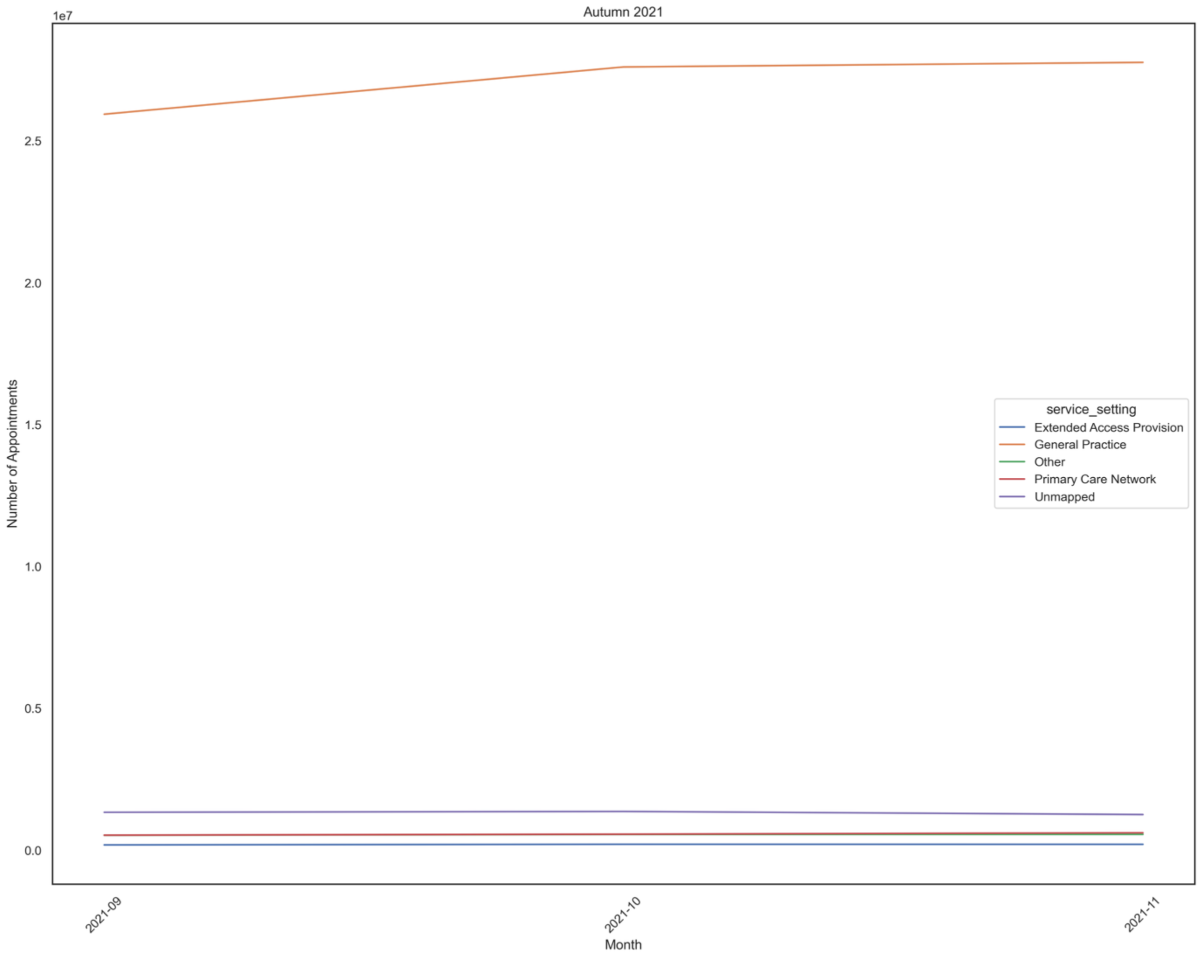
This lineplot shows the distribution of Context Types across the NHS network. Context types are appointments that either involved the patient or not. The chart shows that appointments that do not directly involve the patient are dominant only towards winter but declines and overtaken by direct patient encounters during and after winter. Key insight: from winter and beyond, prefer appointments that directly involve them.

Figure 3.



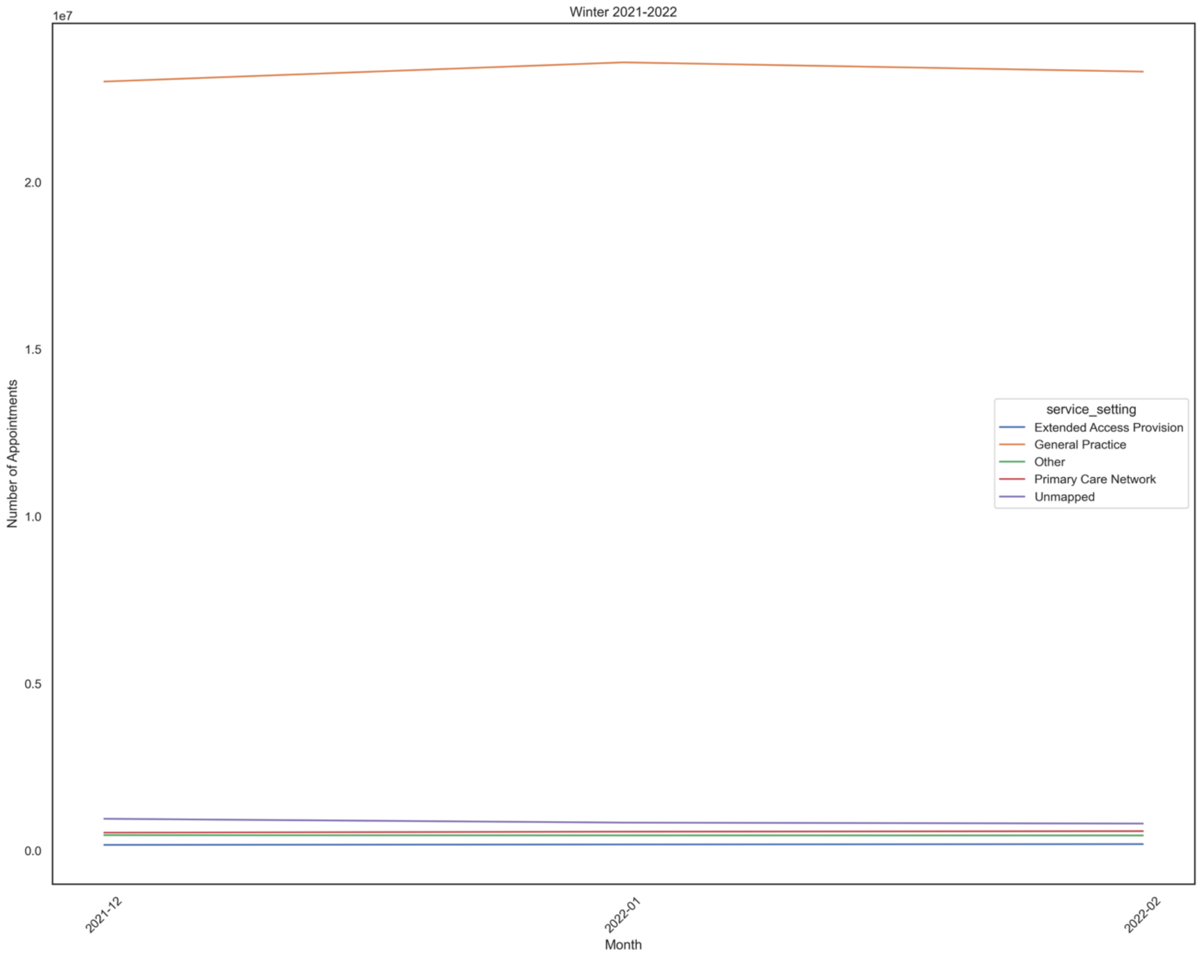
This lineplot shows the distribution of National Category across the reporting period. The key insight is the General Consultation Routine within the Care Related Encounter category dominates appointments.

Figure 4.



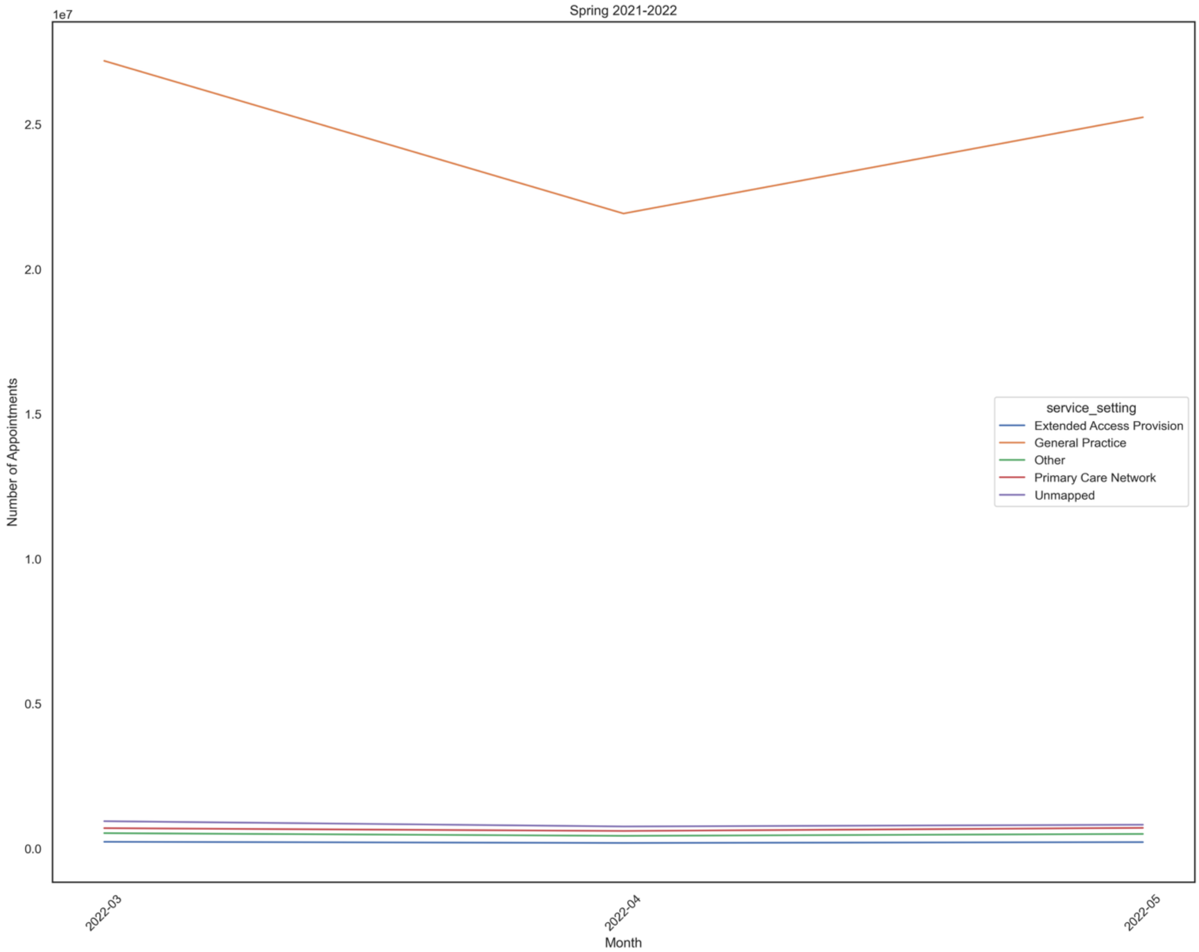
This lineplot shows service settings by season (Autumn). Key insight is that GP appointments tend to increase slightly towards the winter season.

Figure 5.



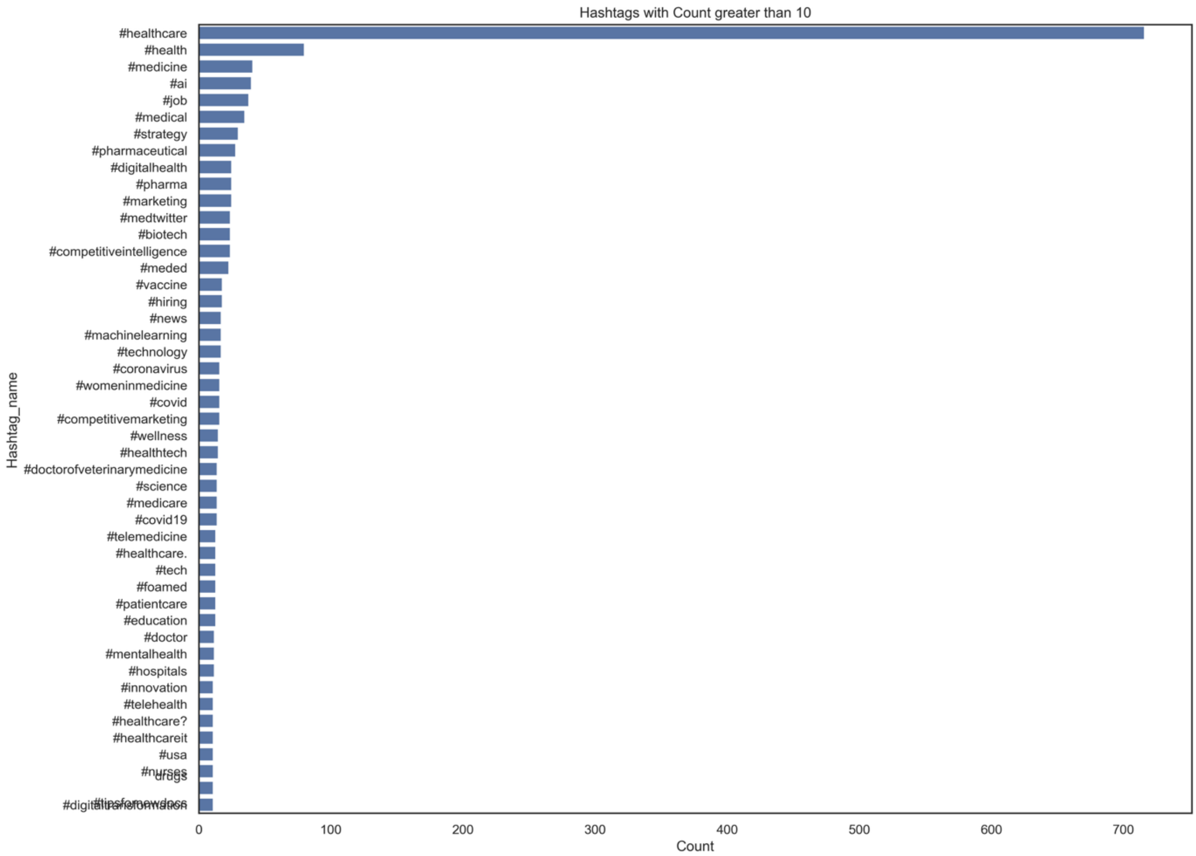
This lineplot shows service setting by season (Winter). Key insight, GP again appointments dominates and peaks in January with slight decline afterwards.

Figure 6.



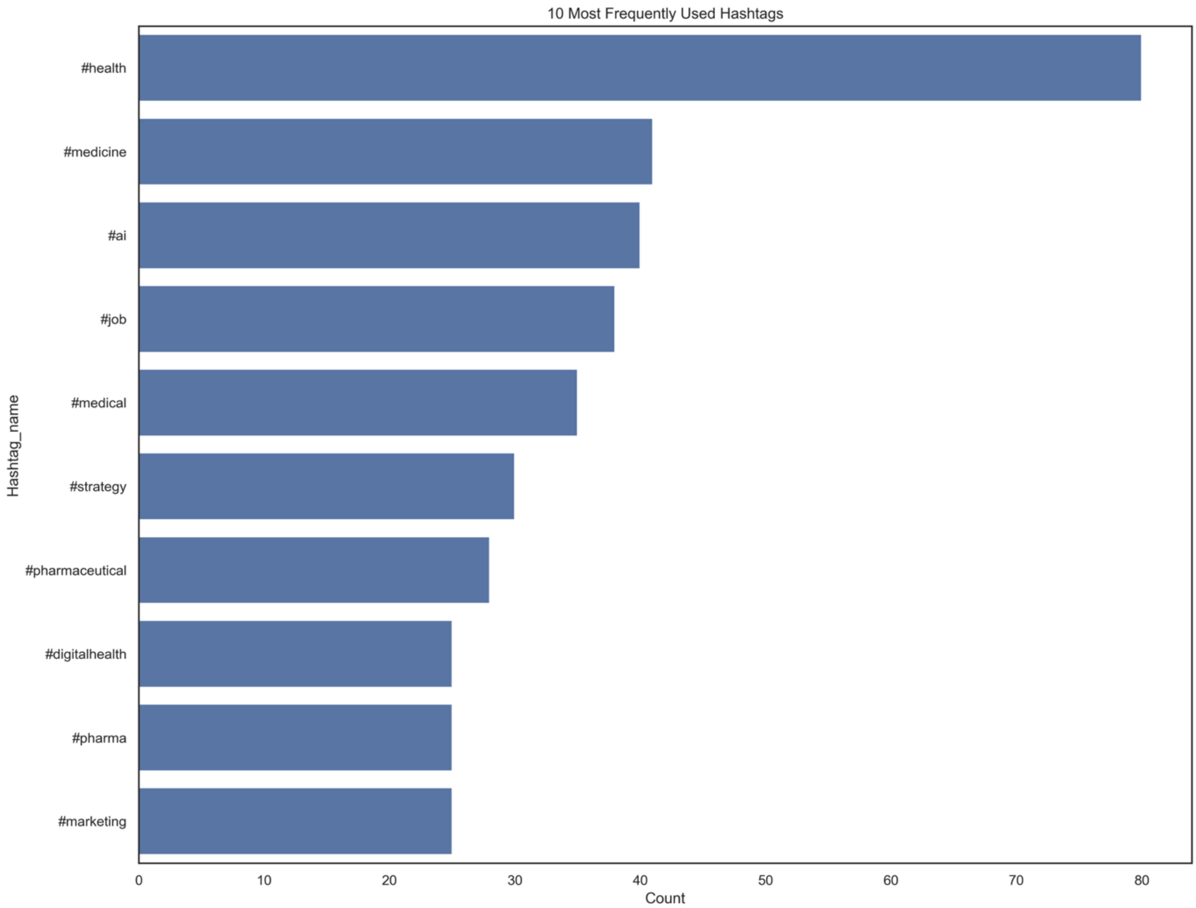
This linepot shows seasonal service setting (Spring). Key insight: GP appointments tend to decline after winter and continues to do so into mid-spring (April) and begins to rise sharply again.

Figure 7.



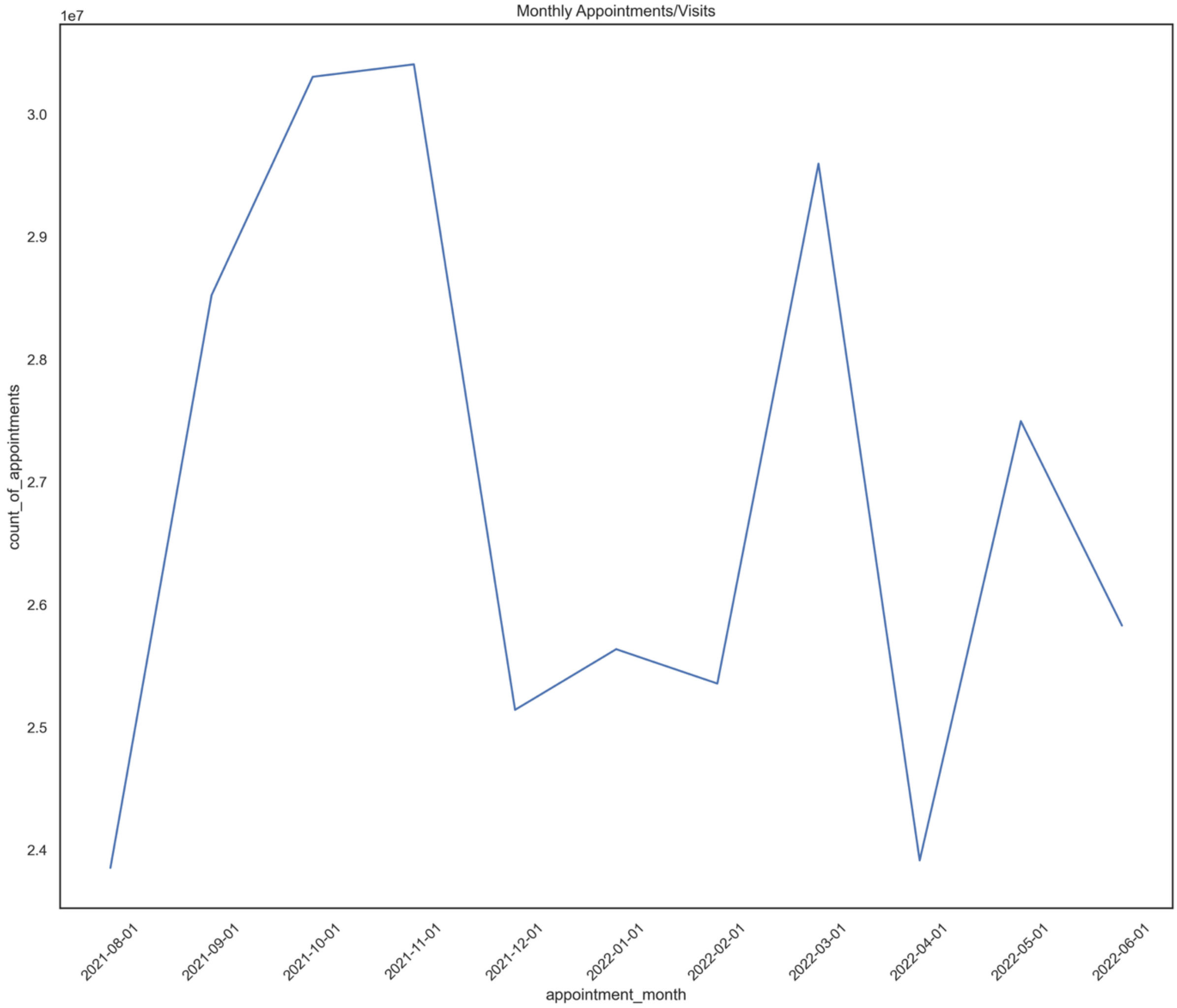
This barplot shows how popular hashtags on Tweeter compare to each other. Key insight: matters relating to healthcare are of great concern to the population.

Figure 8.



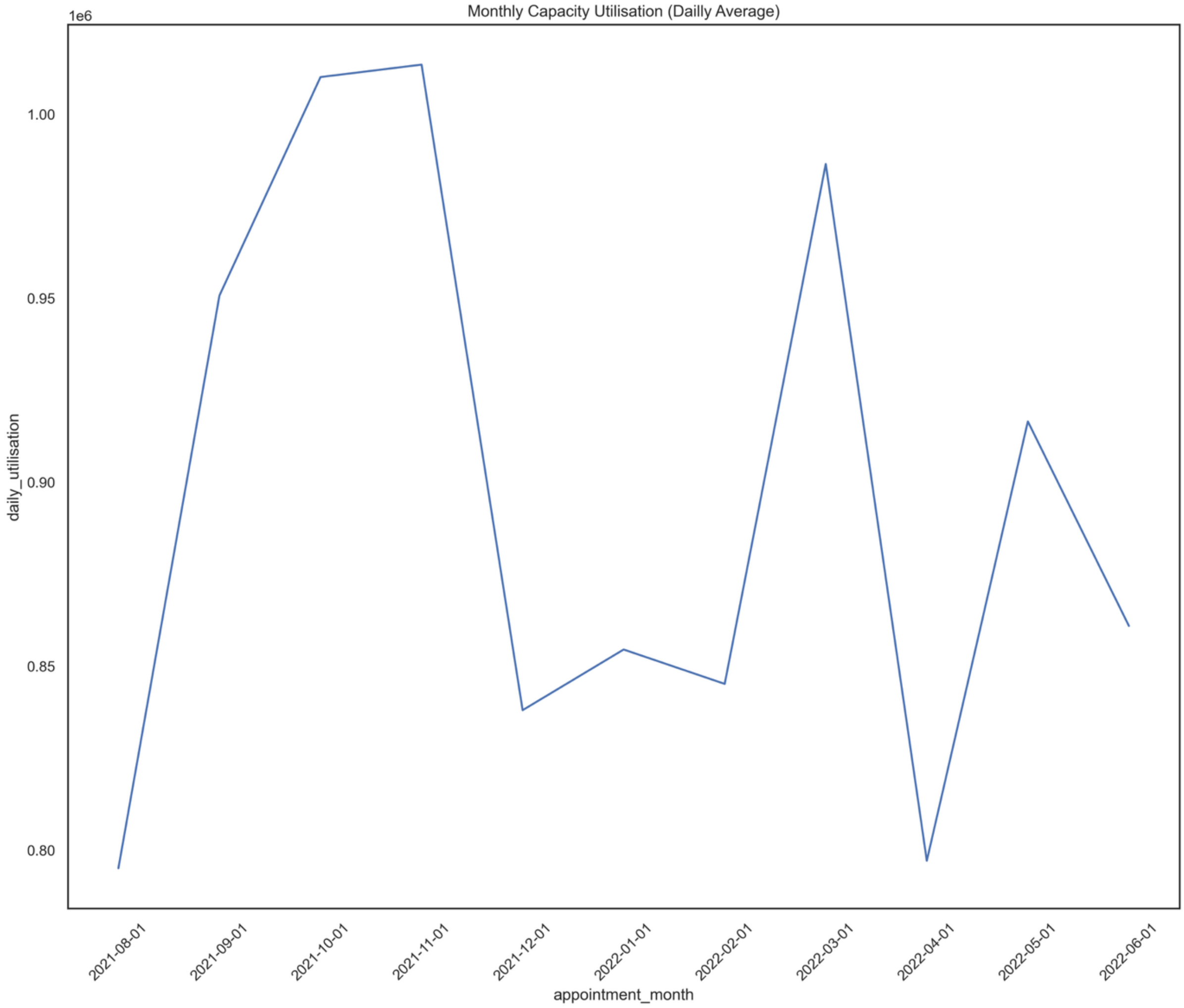
This barplot shows 10 most hashtags with the exclusion of #healthcare. Key insight: even with the exclusion of healthcare, matters relating to health continue to dominate discussion among the population.

Figure 9.



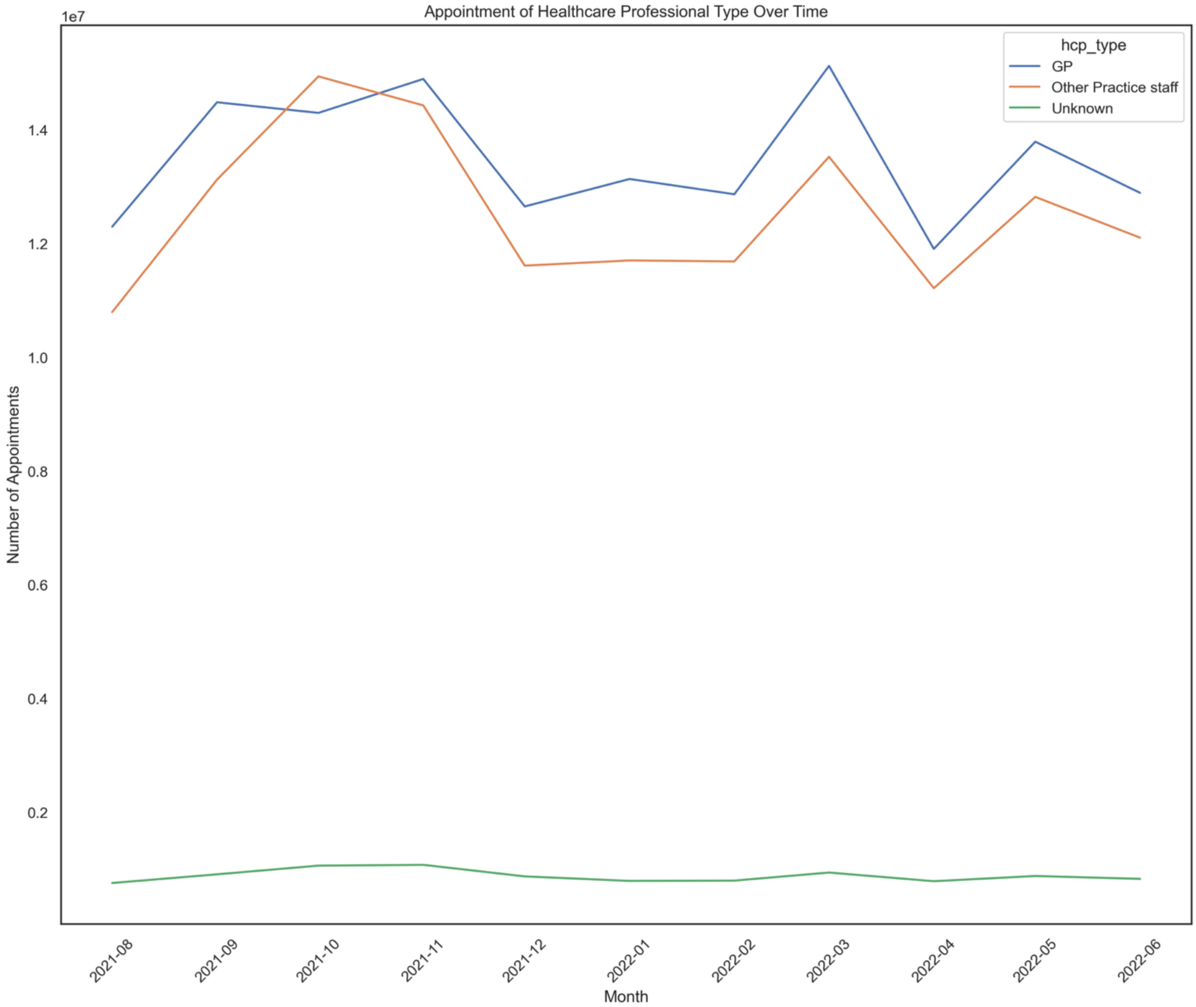
This lineplot shows the distribution of resource utilisation across the period. Key insight: demand remains consistently high (24- 31 million) with no sustained downward trend. The low activity in December is consistent with Christmas holiday period.

Figure 10.



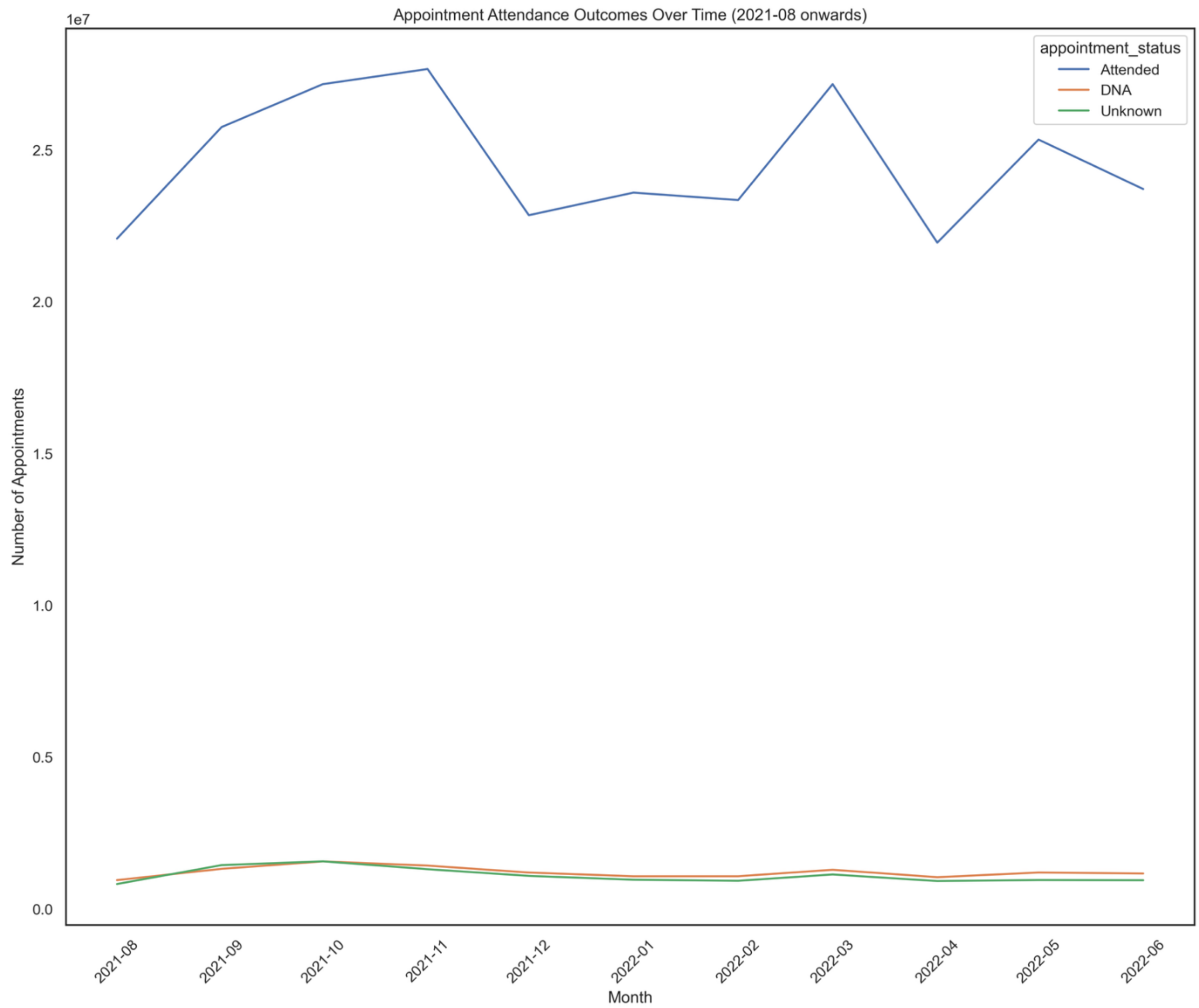
This lineplot shows daily average capacity utilisation. Key insight: daily demand remians high (800,000 – 1m+). There is sustained optimum capacity utilisation compared to the 1.2m NHS benchmark. This mirrors the monthly pattern observed in figure 9 above.

Figure 11.



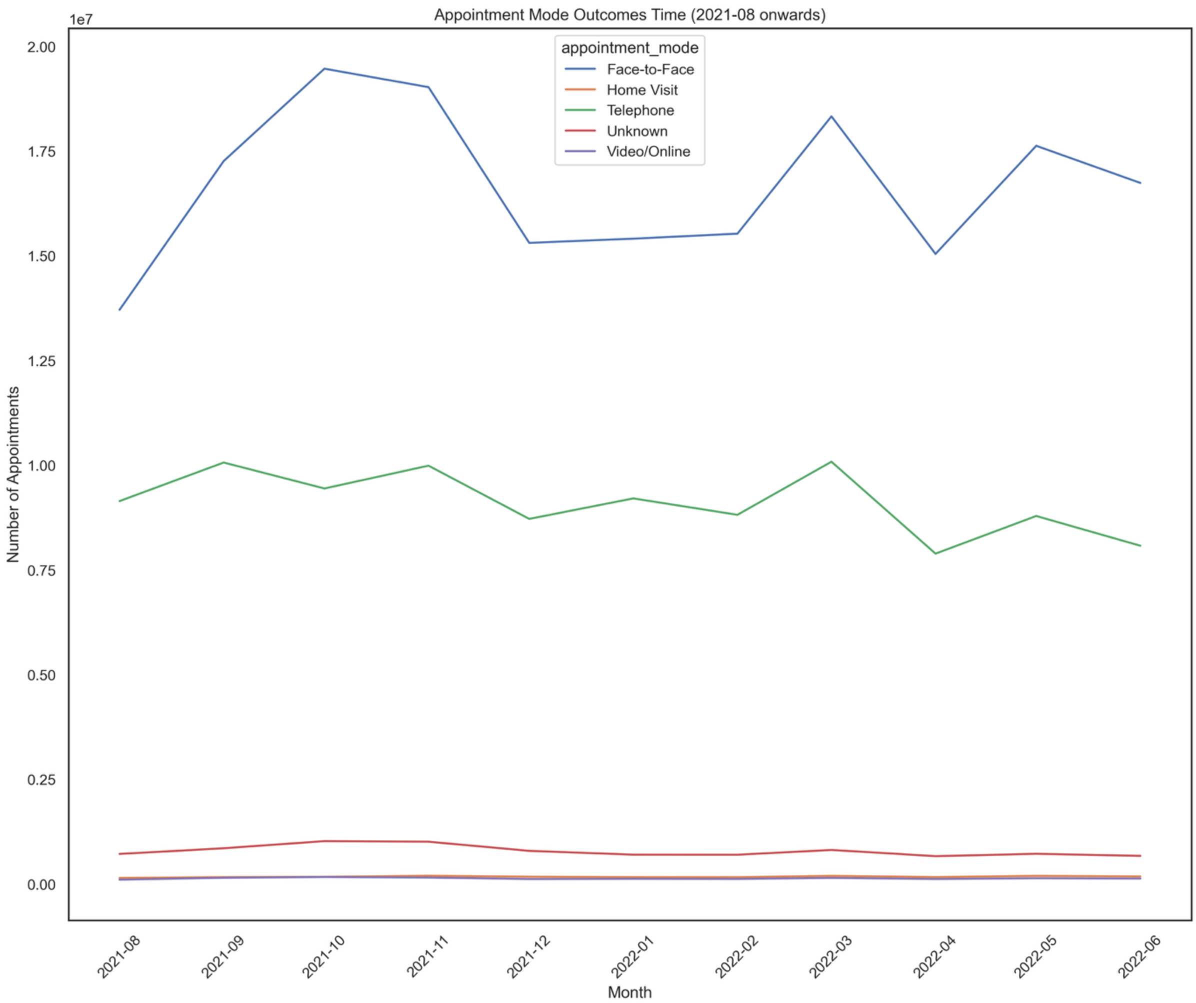
This lineplot shows distribution of appointments by hcp type. Key insight: GP's and other practice staff (including nurses) continue to handle bulk of the appointments (over 1m - 1.4m+ per month).

Figure 12.



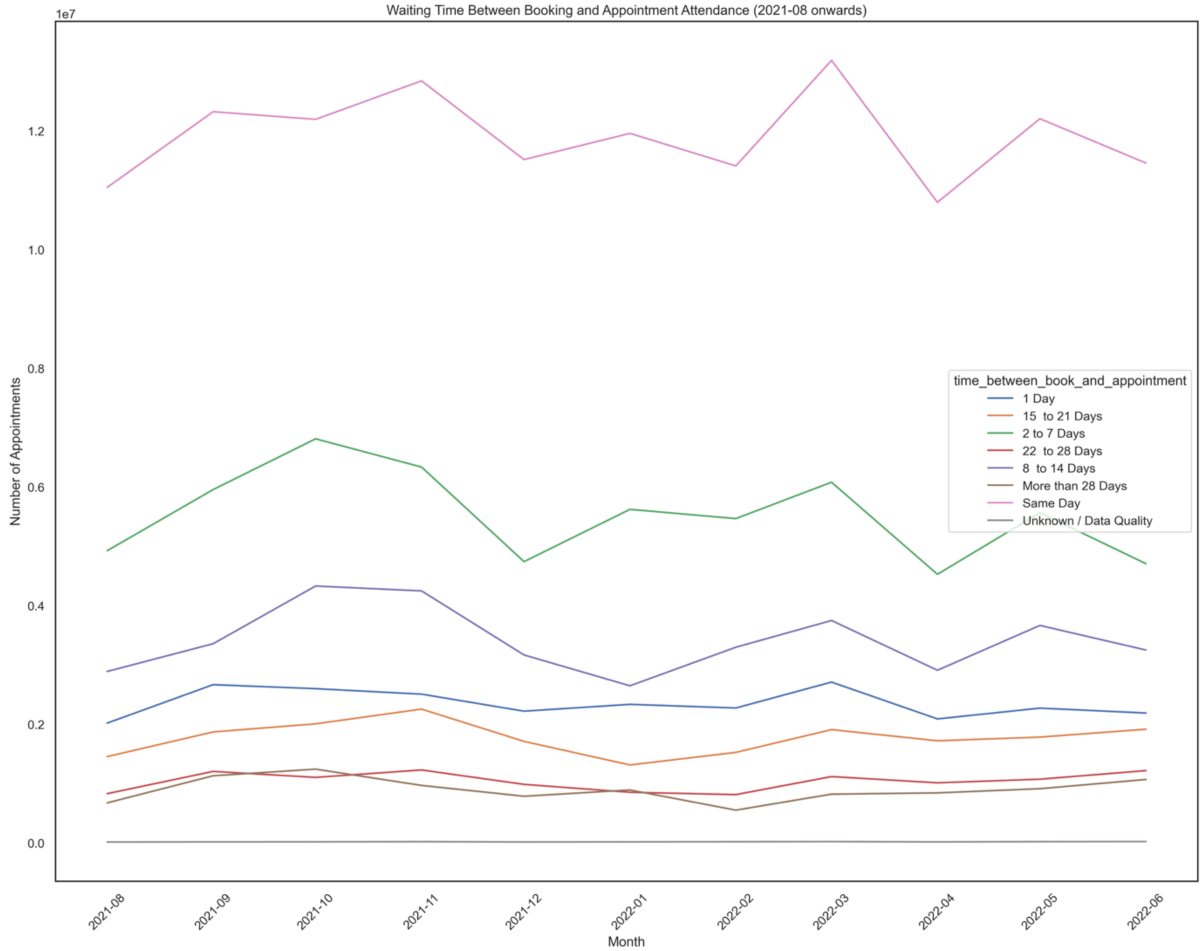
This lineplot shows appointment attendance over the period. Key insight: larger chunk of appointments is attended (between 2.2 – 3m per month). Missed appointments are minimal (approximately 100,000/month).

Figure 13.



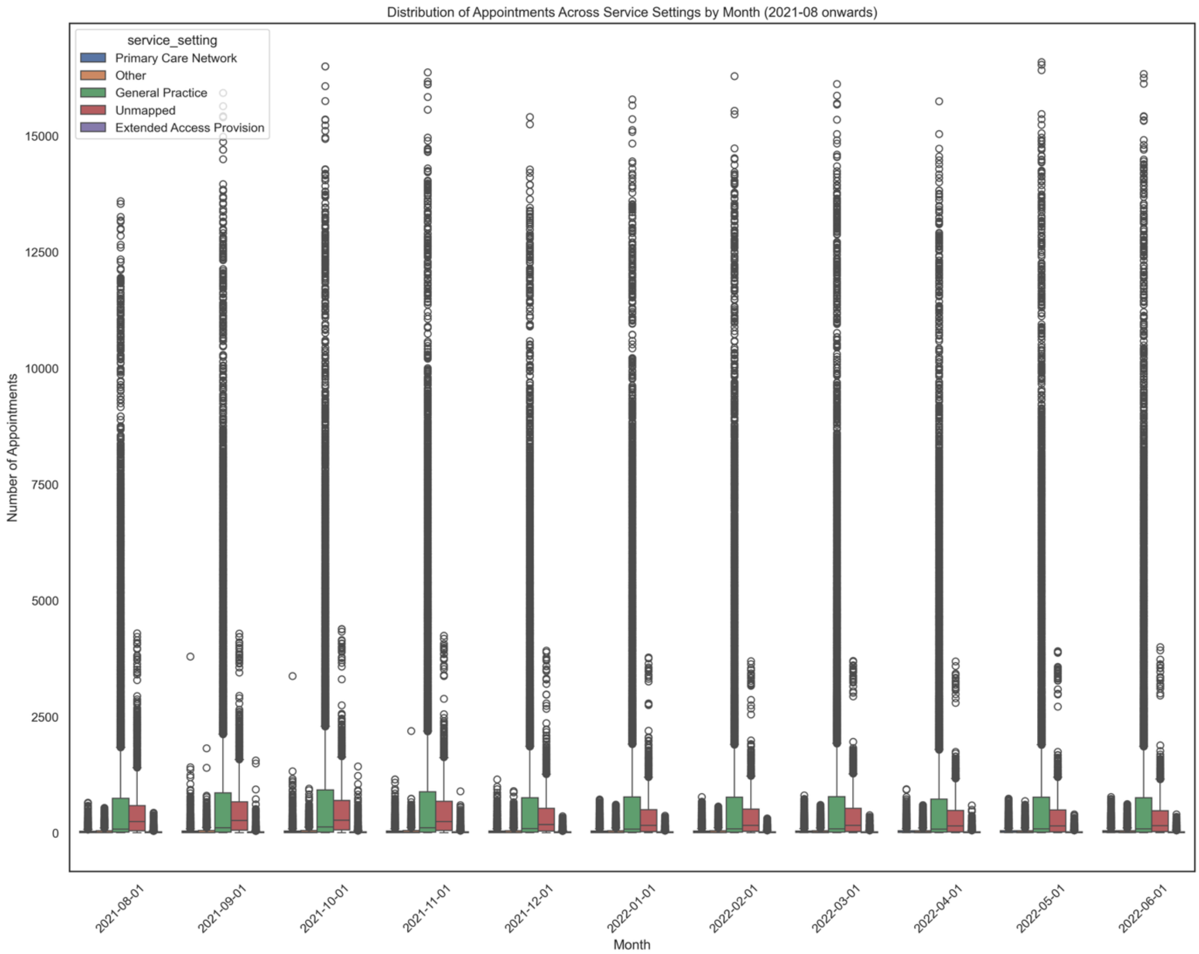
This lineplot shows distribution of appointment mode. Key insight: face to face service delivery continue to dominate (over 1.3m), followed by telephone (a little under 1m) per month.

Figure 14.



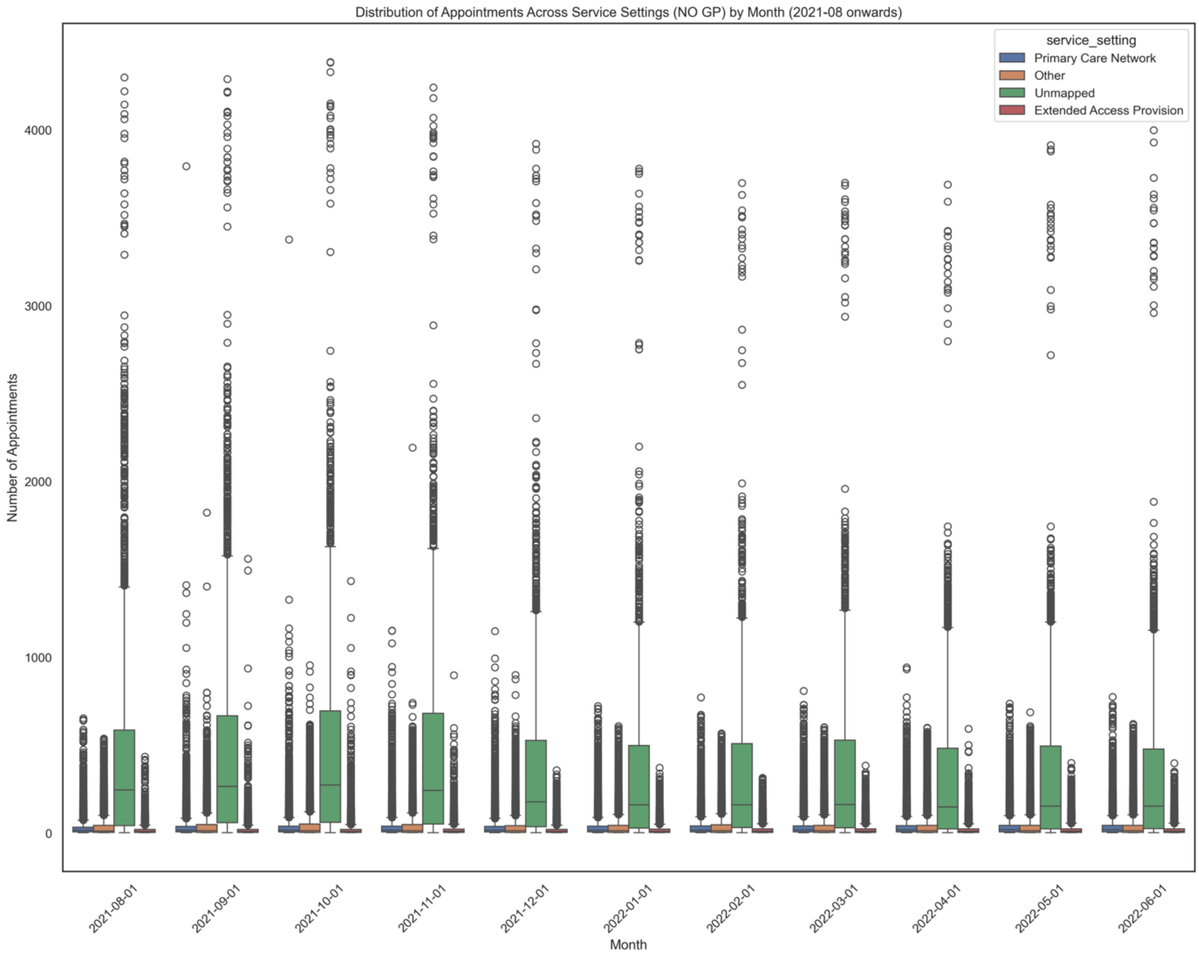
This lineplot shows distribution of waiting times. Key insight: Majority of appointments are delivered same-day, followed by 2-7 days. Most patients (0.5m – 1.2m) are seen either same-day or within a week.

Figure 15.



This boxplot shows distribution of appointments across service settings. Key insight: Primary care demand is overwhelmingly absorbed by GP practices rather than alternative settings. With high variability, workload distribution across GP practices is highly uneven.

Figure 16.



This boxplot shows distribution of service setting without GPs. Key insight: by excluding GP's, Unmapped service setting overwhelmingly absorbed primary care provision. (Unmapped = no record of category against an appointment slot type). There is also high variability within this group.

Patterns and Trends

From the charts, the three (3) broad areas of problems facing the NHS have been investigated and key findings communicated. This section will highlight the prominent patterns and trends as observed.

- On the volume of appointments, there is sustained high utilisation across the period. There were no prolonged upward/downward trends. Demand for appointments remained stable.
- There is evidence of seasonal pattern. This peaks in Oct -Nov 2021 and March 2022, but drops in Dec & April. This seasonal fluctuation is consistent with winter pressure and holiday season effects.
- A pattern of resiliency within the system was observed. There were no long-term decline and the system bounces back quickly after very dip.
- Another pattern observed is high volatility within the GP practices. This could be the result of location population density (urban vrs rural). This observation requires further investigation as the dataset is silent on population metrics.

Conclusion & Recommendations

In response to the problem statement, the following has been suggested based on the analysis outcomes:

- Currently, resource utilisation is optimum. No evidence of severe under-utilisation or collapse of services due to high demand.
- Capacity utilisation is within the NHS guidelines. No compelling evidence to increase staffing levels.
- Most appointments are attended. The few missed appointments require further investigation. The circa 100000/month missed appointments may be considered high under certain conditions
- No evidence of worsening backlog of appointments. Most waiting time are within 1-week.
- Dominant national issue is #healthcare. Policy makers need to pay attention

Limitations

- Major limitation is time. No much time to do a very good job
- Word count. No room to do a detailed work